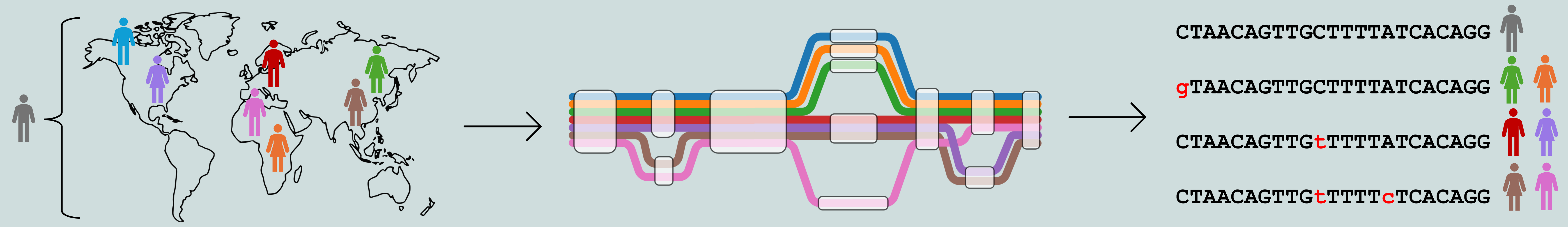
Variant and haplotype-aware guide RNA design toolkit for CRISPR-Cas genome editing: integration of population-based variant data and pangenomics

Department of Computer Science
University of Verona, Italy

Giulia Carone

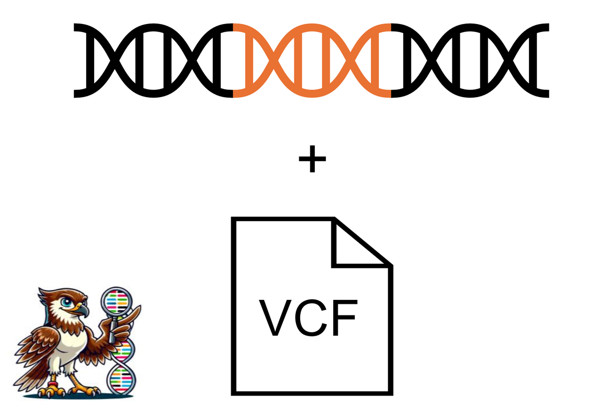
Introduction

Current guide RNA (gRNA) design tools rely primarily on the reference genome, often overlooking individual genetic variation. However, personal variants can significantly impact CRISPR-Cas genome editing by disrupting target sites, creating novel off-targets, and altering Cas protein cleavage efficiency. To improve the reliability of CRISPR-based therapies, it is essential to incorporate individual-specific genomic variation into gRNA design and evaluation. This project aims to develop computational frameworks for designing gRNAs that account for genetic variation derived from large-scale population datasets [1] and pangenome references [2].

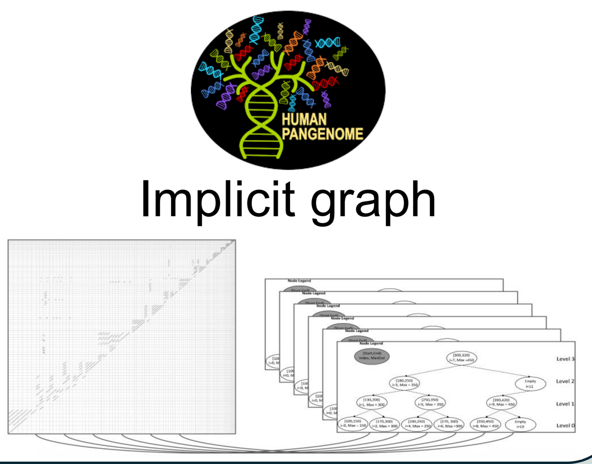


Methodology

Reference target region



HPRC v2.0 Assemblies



Implicit graph

GUIDE SEARCH

...0010000101000100...

CTAACAGTTGCTTTTATCACAGG

Reported spacer

ON AND OFF-TARGET EVALUATION

On-target activity
- Azimuth
- Rule Set 3
- CRISPRon
- PLM-CRISPR
- sgDesigner

Variant impact on guide efficiency

CFD

Guide specificity

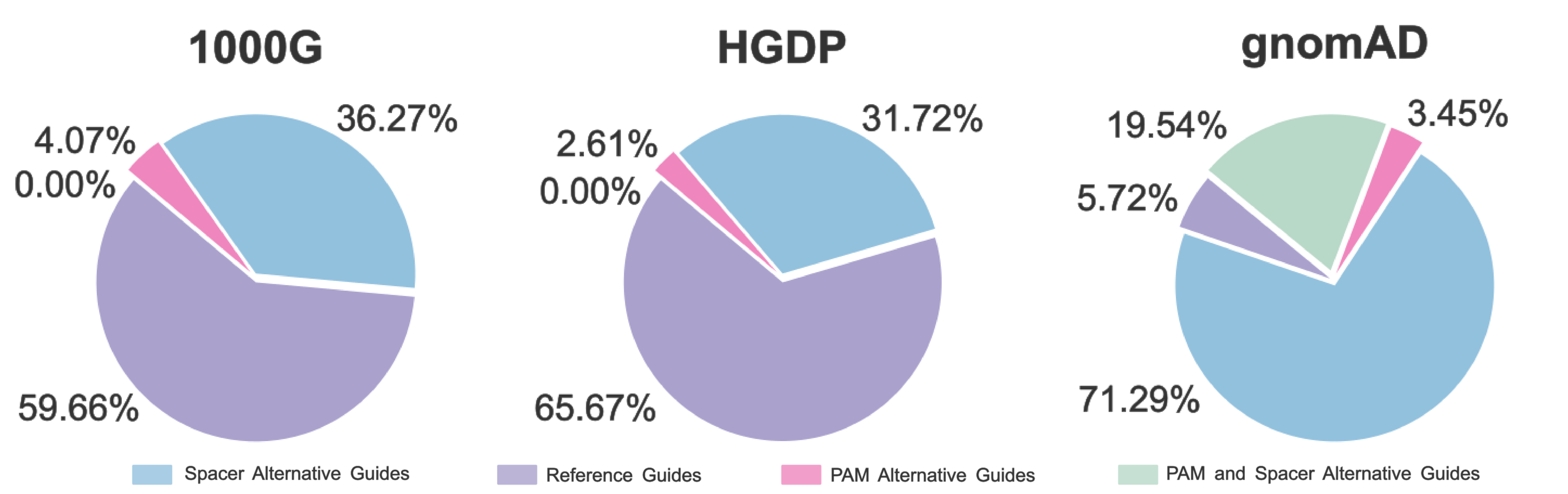
CRISPRitz

CRISPRme

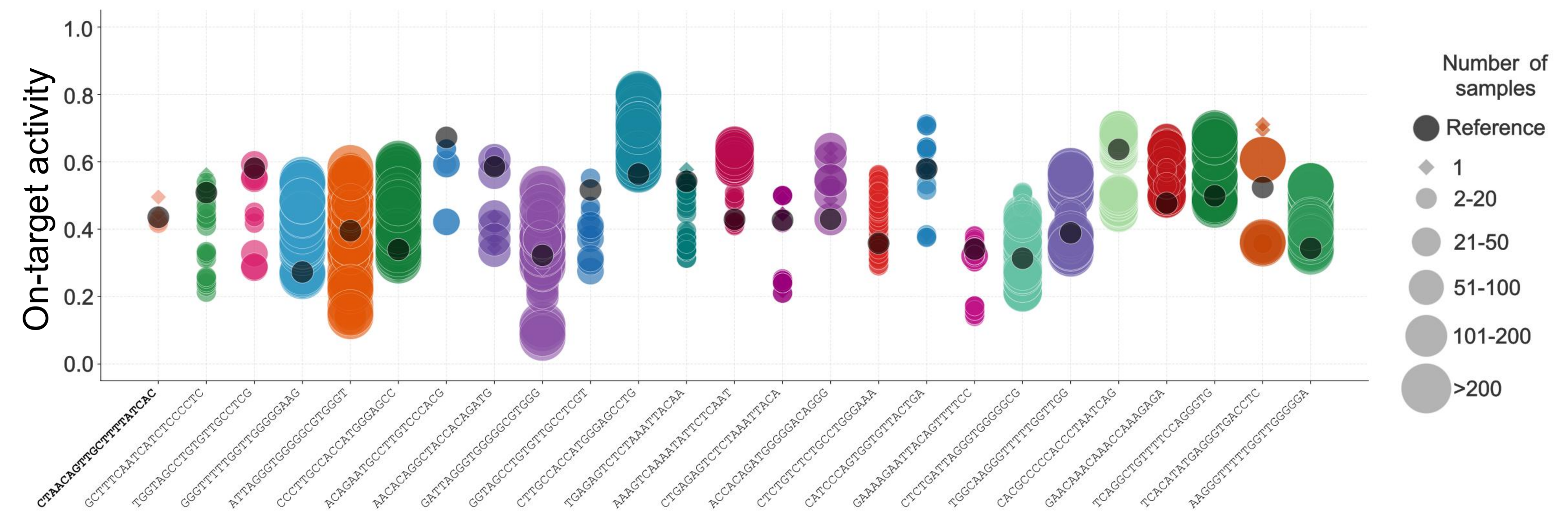
Results

CRISPR-HAWK

BCL11A Enhancer – gRNA classification



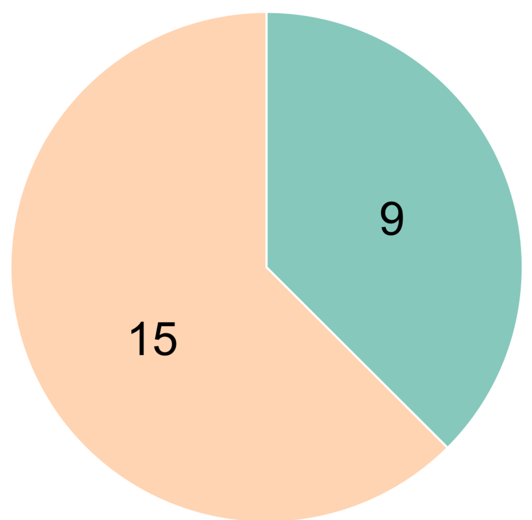
On-target activity (Azimuth)



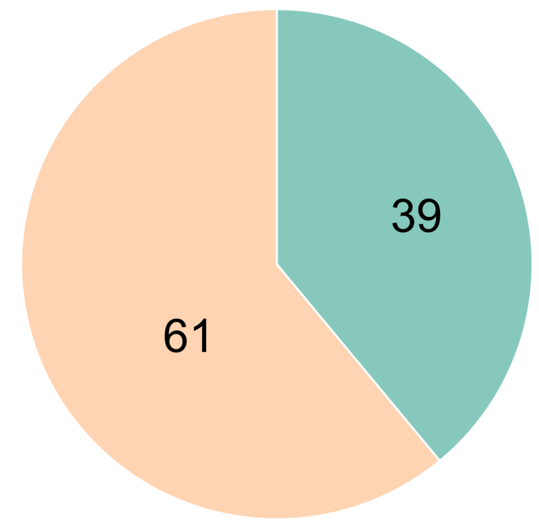
CRISPR-HAWK PANGENOME

New guides found in HPRC data

BCL11A Enhancer

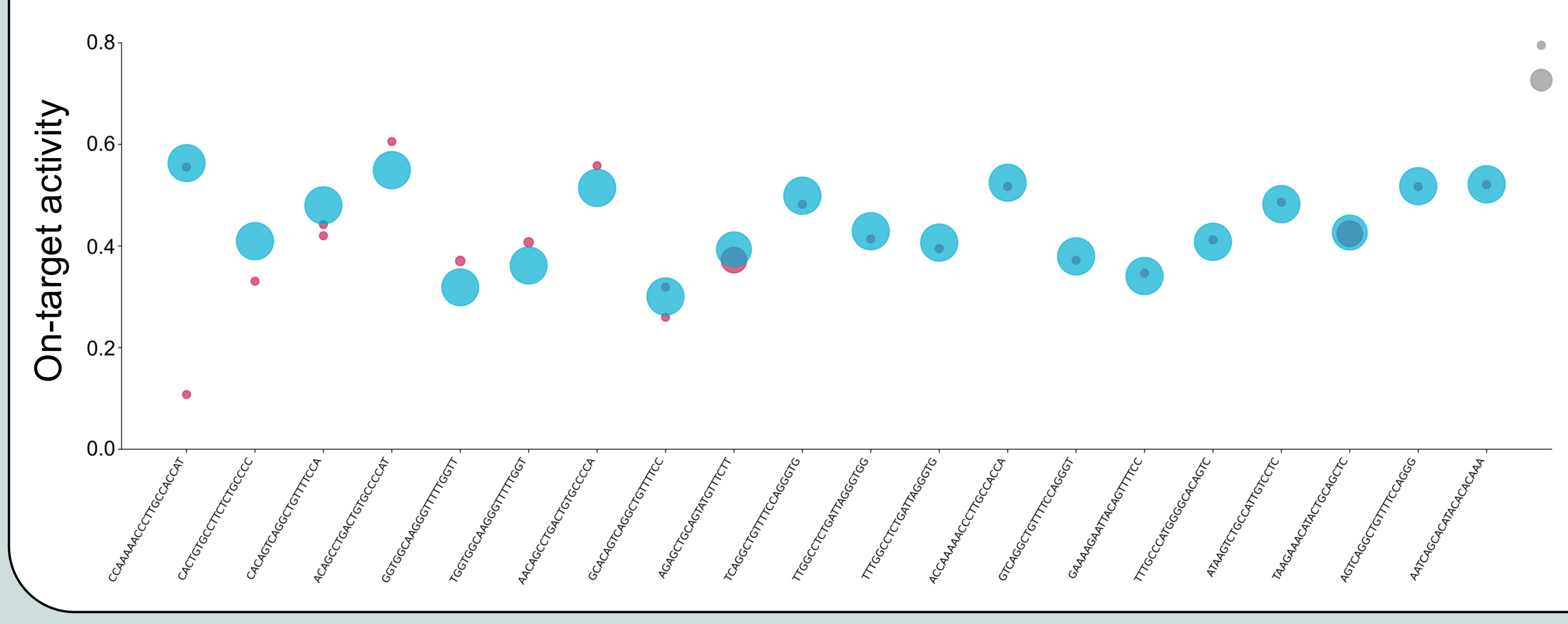


Different scores
Different guides



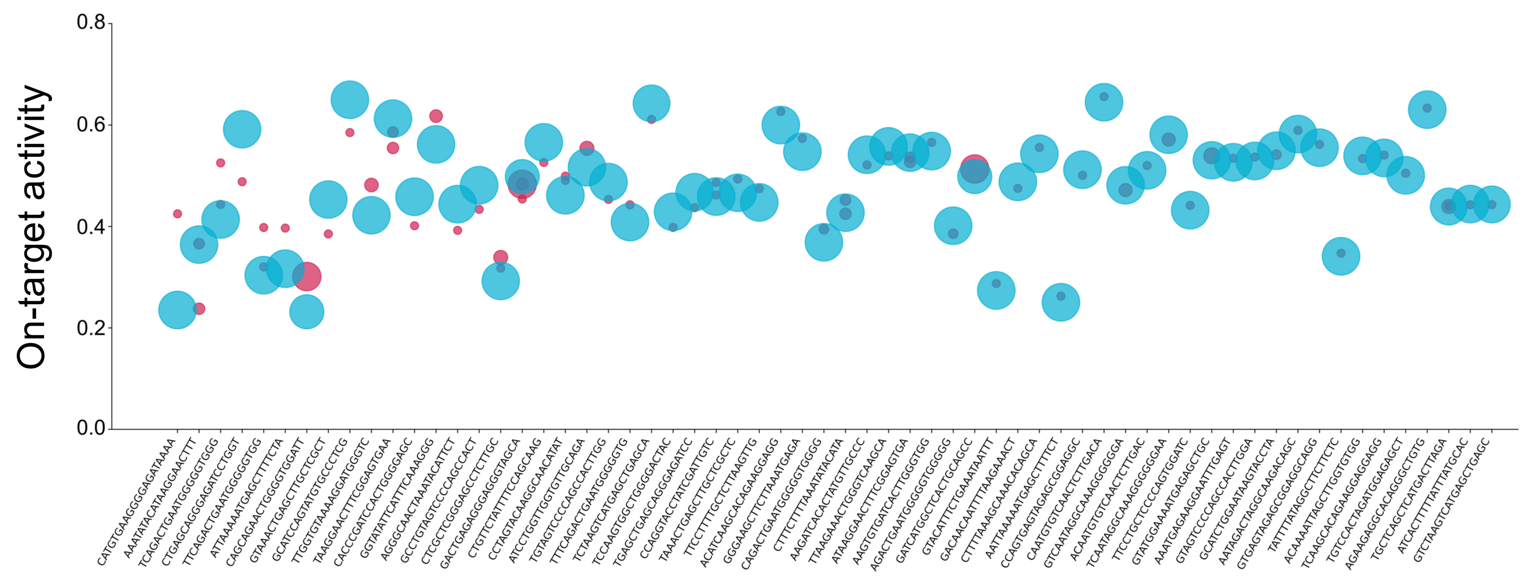
CCR5

On-target activity (Azimuth)



Number of samples
1 sample
50 samples
100 samples
200 samples

Guide type
Alternative
Reference



Conclusions

Our results demonstrate that naturally occurring variants can significantly impact guide performance, highlighting the importance of accounting for population-based genetic diversity in genome editing strategies. Furthermore, the use of high-quality genomic resources, such as HPRC assemblies, combined with pangenome graph frameworks, enabled the discovery of novel variant-harbouring guides and the characterization of additional genetic contexts affecting guide efficiency.

References

[1] A. Kumbara, G. Carone et al., bioRxiv 2025
[2] W. Liao, M. Asri et al., Nature 2023

giulia.carone@univr.it